Pythagoras' Theorem — squares on the sides of a right angle

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Worksheet

Part A

1. a) **169** b) **24** ($24^2 = 576$) c) **0.7** ($0.7^2 = 0.49$)
2. $\frac{5^9}{5^6} = 5^3 =$ **125**
3. **9 and 10** ($81 < 90 < 100$). 90 is 9 away from 81 but 10 away from 100, so it is slightly **closer to 9** ($\sqrt{90} \approx 9.49$).
4. $\sqrt{40}$ **and** π**.** ($\sqrt{36} = 6$, $\sqrt{9/16} = \frac{3}{4}$, and $\frac{2}{7}$, $0.\overline{3} = \frac{1}{3}$ are all rational.)
5. $3x + 90 = 180 \Rightarrow x =$ **30**
6. **RHS** (right angle–hypotenuse–side) — today's topic is the reason this test works: two sides of a right-angled triangle fix the third.
7. $m = \frac{11-3}{3-(-1)} = \frac{8}{4} =$ **2**
8. $9a^2 + 16a^2 = 25a^2 = (5a)^2$. The sum of two squares is itself a perfect square — it is the 3–4–5 triple (scaled by a). A deliberate set-up for Part B.

C1

1. a) **PR** b) **LM** c) **BC** (the side not touching the right-angle vertex).

2.

a) $x^2 = 9^2 + 12^2 = 81 + 144 = 225$, $x =$ **15** (3–4–5 × 3)

b) $x^2 = 20^2 + 21^2 = 400 + 441 = 841$, $x =$ **29**

c) 26 is the hypotenuse: $x^2 = 26^2 - 10^2 = 676 - 100 = 576$, $x =$ **24** (5–12–13 × 2)

d) 17 is the hypotenuse: $x^2 = 289 - 225 = 64$, $x =$ **8**

e) $x^2 = 2^2 + 1.5^2 = 4 + 2.25 = 6.25$, $x =$ **2.5 m** (3–4–5 × 0.5)

f) 41 is the hypotenuse: $x^2 = 1681 - 81 = 1600$, $x =$ **40**

3.

	Working	Exact	🖩 2 d.p.
a)	$1 + 1 = 2$	$\sqrt{2}$ cm	1.41 cm
b)	$4 + 9 = 13$	$\sqrt{13}$ cm	3.61 cm
c)	$49 - 16 = 33$	$\sqrt{33}$ cm	5.74 cm
d)	$100 - 25 = 75$	$\sqrt{75}$ cm ($= 5\sqrt{3}$, see F2)	8.66 cm

C2

4. a) $289 - 64 =$ **225 cm²**. b) $\sqrt{64} = 8$, $\sqrt{225} = 15$, $\sqrt{289} = 17$ → sides **8 cm, 15 cm, 17 cm**.
c) The area of the square on the hypotenuse equals the total area of the squares on the two shorter sides. c^2 is literally the area of a square with side c, so the theorem says "big square = two small squares added together" (and you could cut up the two small squares to exactly cover the big one).

5.

Triangle	$a^2 + b^2$	c^2	Result
A (7, 24, 25)	$49 + 576 = 625$	625	**right-angled**, opposite the 25 side
B (6, 8, 11)	$36 + 64 = 100$	121	c^2 bigger → **obtuse**
C (9, 12, 14)	$81 + 144 = 225$	196	c^2 smaller → **acute**
D (1.2, 3.5, 3.7)	$1.44 + 12.25 = 13.69$	13.69	**right-angled**, opposite the 3.7 m side
E (10, 10, 14)	$100 + 100 = 200$	196	**acute** (just!)

E: $200 \neq 196$, so the largest angle is slightly less than 90° (about 88.9°). Diagrams are not to scale and measurements by eye can't tell a 1° difference — only the calculation proves it.

6. Jordan treated 13 as a shorter side and **added**. The hypotenuse is the longest side, so x must be *less than* 13 — an answer of 13.9 is impossible. Correct: $x^2 = 13^2 - 5^2 = 169 - 25 = 144$, $x =$ **12 cm**.

7. a) $20^2 + 48^2 = 400 + 2304 = 2704 = 52^2$ ✔. It is **(5, 12, 13) × 4**.
b)

n	3	5	7	9	11
$\frac{n^2-1}{2}$	4	12	24	40	60
$\frac{n^2+1}{2}$	5	13	25	41	61

c) $9^2 + 40^2 = 81 + 1600 = 1681 = 41^2$ ✔. The two larger numbers are always **consecutive** (they differ by 1).

C3

8. $60^2 + 80^2 = 3600 + 6400 = 10\,000 = 100^2$, so a square corner would give exactly 100 cm.
$101^2 = 10\,201 > 10\,000$, so the angle is **obtuse — slightly more than** 90°. The corner is not square.

9.
a) $h^2 = 6.5^2 - 2.5^2 = 42.25 - 6.25 = 36$, $h =$ **6 m** (2.5, 6, 6.5 = (5, 12, 13) × 0.5).
b) New height $= \sqrt{6.5^2 - 3.5^2} = \sqrt{42.25 - 12.25} = \sqrt{30} \approx 5.477$ m. Drop $= 6 - 5.477 \approx$ **0.52 m (52 cm)** — **less** than the 1 m the foot moved. (The relationship isn't linear.)
c) $h^2 + (\frac{h}{4})^2 = 6.5^2 \Rightarrow h^2 + \frac{h^2}{16} = 42.25 \Rightarrow \frac{17h^2}{16} = 42.25 \Rightarrow h^2 = \frac{16 \times 42.25}{17} \approx 39.765$, $h \approx$ **6.31 m** (foot about 1.58 m out). Exact: $h = \frac{26}{\sqrt{17}}$.

10.
a) $A(-3, -1)$, $B(5, 5)$, $C(1, 7)$
b) Horizontal leg $5 - (-3) = 8$, vertical leg $5 - (-1) = 6$. $AB^2 = 64 + 36 = 100$, **AB = 10**.
c) AC: across $1 - (-3) = 4$, up $7 - (-1) = 8$ → $AC^2 = 16 + 64 = 80$, **AC =** $\sqrt{80}$ $(= 4\sqrt{5})$.
BC: across 4, up 2 → $BC^2 = 16 + 4 = 20$, **BC =** $\sqrt{20}$ $(= 2\sqrt{5})$.
d) $AC^2 + BC^2 = 80 + 20 = 100 = AB^2$ ✔ → right-angled, with the right angle **at C** (opposite the longest side AB).

C4 Connections

X1. Drop a vertical from the top-right corner: it cuts off a right-angled triangle with base $21 - 9 = 12$ cm and hypotenuse 13 cm. $h^2 = 169 - 144 = 25$, $h = 5$ **cm**.
Area $= \frac{1}{2}(9 + 21) \times 5 =$ **75 cm²**. Perimeter $= 9 + 21 + 13 + 5 =$ **48 cm**.

X2. $x^2 + (x + 7)^2 = (x + 8)^2$
$x^2 + x^2 + 14x + 49 = x^2 + 16x + 64$
$x^2 - 2x - 15 = 0$. Try values: $x = 5$: $25 - 10 - 15 = 0$ ✔ (or factorise: $(x - 5)(x + 3) = 0$, reject $x = -3$ as a length can't be negative).
$x = 5$: sides 5, 12, 13 cm. Area $= \frac{1}{2} \times 5 \times 12 =$ **30 cm²**; perimeter $= 5 + 12 + 13 =$ **30 cm**. The area and perimeter have the **same number** (but different units, so they aren't really "equal" — a good discussion point).

X3. Half the base is 1.6 m. Height $= \sqrt{2^2 - 1.6^2} = \sqrt{4 - 2.56} = \sqrt{1.44} =$ **1.2 m**.
Cross-section area $= \frac{1}{2} \times 3.2 \times 1.2 =$ **1.92 m²**. Volume $= 1.92 \times 5 =$ **9.6 m³** $= 9.6 \times 1000 =$ **9600 L** (1 m³ = 1000 L).

X4. a) $m_{AB} = \frac{6}{8} = \frac{3}{4}$. Gradient is a *ratio* (rise per run), not a length: e.g. $(-3, -1)$ to $(1, 2)$ also has gradient $\frac{3}{4}$ but length 5. You need the actual rise and run, then Pythagoras.
b) $m_{AC} = \frac{8}{4} = 2$, $m_{BC} = \frac{7 - 5}{1 - 5} = -\frac{1}{2}$. Product $= -1$. AC and BC meet at a right angle (Q10 d) — **perpendicular lines have gradients multiplying to −1** (a Year 10 result they've just discovered).

X5. a) $\sqrt{36 + 64} =$ **10 cm**.
b) New sides 7.5 cm and 10 cm: $\sqrt{56.25 + 100} = \sqrt{156.25} =$ **12.5 cm → 25% increase** (every length scales by 1.25). Area: 48 → 75 cm², $\frac{27}{48} =$ **56.25% increase** ($1.25^2 = 1.5625$, link to Week 12 scale factors).
c) New sides 9 cm and 8 cm: $\sqrt{81 + 64} = \sqrt{145} \approx 12.04$ cm → $\approx$ **20.4% increase**. Only one of the two squares being added grows, so the diagonal grows by less than 50% (the diagonal isn't proportional to one side on its own).

Part D

11. $d = \sqrt{30.5^2 + 15.25^2} = \sqrt{930.25 + 232.5625} = \sqrt{1162.8125} \approx$ **34.10 m**. Two sides $= 45.75$ m, so the diagonal is about **11.65 m shorter**.

12. a) $0.7 \times 14 =$ **9.8 m**. b) $\sqrt{9.8^2 + 0.7^2} = \sqrt{96.04 + 0.49} = \sqrt{96.53} \approx$ **9.82 m**. c) $\frac{0.7}{9.8} = \frac{1}{14}$.
Note: the sloping length is only about 2.5 cm longer than the run — for gentle slopes, length ≈ run.

13. a) $\sqrt{7^2 + 24^2} = \sqrt{625} =$ **25 km**. b) Now 2 km north and 24 km east: $\sqrt{4 + 576} = \sqrt{580} \approx$ **24.08 km**. c) 36 km is the **distance travelled** along the path; the straight-line distance from the marina is the hypotenuse, which is always shorter than the sum of the legs (unless the yacht sails in one straight line).

Part E

14. a) Arrangement 1: one tilted square, area c^2. Arrangement 2: two squares, total area $a^2 + b^2$.
b) Both big squares have the same area $(a + b)^2$, and both contain the same four triangles. Taking the same triangles away from the same big square must leave the same white area, so $c^2 = a^2 + b^2$.
Nothing depended on particular values of a and b, so it works for every right-angled triangle. *(Check the inner shape in Arrangement 1 really is a square: each corner angle is 180° minus the two acute angles of a triangle, which sum to 90° — Week 13 angle sum.)*
c) $(a + b)^2 = c^2 + 4 \times \frac{1}{2}ab$
$a^2 + 2ab + b^2 = c^2 + 2ab$
$a^2 + b^2 = c^2$ ∎

15. Base diagonal: $\sqrt{12^2 + 4^2} = \sqrt{160}$ ($= 4\sqrt{10} \approx 12.65$ cm). Space diagonal: $\sqrt{160 + 3^2} = \sqrt{169} =$ **13 cm**. Formula: $d = \sqrt{a^2 + b^2 + c^2}$ (two uses of Pythagoras; the base diagonal squared is $a^2 + b^2$, no need to square-root it in between).

16. a) (2, 1): **3, 4, 5**; (3, 2): **5, 12, 13**; (4, 1): **15, 8, 17**; (5, 2): **21, 20, 29**.
b) $(m^2 - n^2)^2 + (2mn)^2 = m^4 - 2m^2n^2 + n^4 + 4m^2n^2 = m^4 + 2m^2n^2 + n^4 = (m^2 + n^2)^2$ ∎

Homework

1. a) $\sqrt{256 + 900} = \sqrt{1156} =$ **34 cm** (8–15–17 × 2) b) $\sqrt{625 - 49} = \sqrt{576} =$ **24 cm** c) $\sqrt{16 + 36} = \sqrt{52}$ **cm** ($= 2\sqrt{13}$) d) $\sqrt{81 - 9} = \sqrt{72}$ **cm** ($= 6\sqrt{2}$)

2. c) **7.2 cm** d) **8.5 cm**

3. a) $121 + 3600 = 3721 = 61^2$ → **right-angled** b) $64 + 196 = 260 > 256 = 16^2$ → **acute** c) $2.56 + 9 = 11.56 = 3.4^2$ → **right-angled**

4. $\sqrt{3.6^2 + 1.5^2} = \sqrt{12.96 + 2.25} = \sqrt{15.21} =$ **3.9 m** ($0.3 \times (5, 12, 13)$: $1.5, 3.6, 3.9$).

5. a) Across $4 - (-2) = 6$, down $3 - (-5) = 8$: $PQ = \sqrt{36 + 64} =$ **10** b) $m = \dfrac{-5 - 3}{4 - (-2)} = \dfrac{-8}{6} = -\dfrac{4}{3}$

6. a) Vertical part of string $= \sqrt{50^2 - 30^2} = \sqrt{1600} = 40$ m; height $= 40 + 1.5 =$ **41.5 m**. b) $\sqrt{50^2 - 40^2} = 30$ m → height 31.5 m → dropped **10 m**. *Common error:* forgetting the 1.5 m hand height (it cancels in part b).

7. a) $2^4 =$ **16** b) $240 \times 1.15 =$ **$276** c) $3x - 12 = x + 10 \Rightarrow 2x = 22 \Rightarrow x =$ **11** d) total $= 32$, so $x = 32 - 20 =$ **12** e) primes 2, 3, 5 → $\frac{3}{6} = \frac{1}{2}$

8. a) Answers vary. A square corner gives 50 cm; more than 50 cm → obtuse; less → acute.
b) Marking notes: a strong answer says the 50 cm diagonal is the side opposite the corner (the would-be **hypotenuse**), that $30^2 + 40^2 = 50^2$, and that by the **converse** a triangle whose sides fit $a^2 + b^2 = c^2$ must be right-angled — so measuring 50 cm proves the corner is square.

Teacher notes

- **Common misconceptions:**
  - Adding the squares when finding a shorter side (Q6). Habit to build: *"Is my answer shorter than the hypotenuse?"*
  - $\sqrt{a^2 + b^2} = a + b$ (e.g. $\sqrt{9 + 16} = 7$). Counter-example: 3–4–5.
  - Squaring decimals: $1.5^2 = 2.25$, not 2.5 or 3.
  - Rounding the base diagonal before finding a space diagonal (Q15) — keep $\sqrt{160}$ or 160 exactly.
  - Using the converse on the wrong side: c must be the **longest** side.
  - Thinking "looks right" counts (triangle E in Q5).
- **If the student finishes early:** Fast-finisher F9–F12; ask them to prove the converse (construct a right-angled triangle with legs a, b; its hypotenuse must be c; then SSS congruence, Week 14). Euclid's formula (Q16) generates *all* primitive triples when m, n are coprime and not both odd — a good research prompt.
- **If the student struggles:** use squared paper — draw the 3–4–5 triangle with squares on the sides and count unit squares; restrict to Pythagorean triples before surds; label c first every time.
- **Calculator use:** Parts A–C are non-calculator except items marked 🖩; Part D is calculator. Exact (surd) answers are expected whenever no rounding is requested.

- **Facts and contexts:**
  - Netball court 30.5 m × 15.25 m: the standard court dimensions in the World Netball rules.
  - The "1 in 14" ramp rule is presented as hypothetical ("suppose"). For reference, 1:14 is the commonly cited maximum gradient for an accessible ramp under Australian Standard AS 1428.1 — check the current standard before quoting it as fact.
  - The ladder "1 out for every 4 up" (≈75°) rule of thumb appears widely in Australian workplace-safety guidance; it is phrased as a rule of thumb, not a regulation.
  - Ladder, builder, yacht, tent, kite and gate measurements are invented for the exercise.
  - The 3–4–5 "rope-stretcher" story (homework Q8) is kept deliberately vague ("builders and surveyors have long used…"), because claims about ancient Egyptian use are not well documented.
- **Next week (Week 32) builds on:** distance between points on the Cartesian plane (Q10, X4, F10) → the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ and midpoints; 3D diagonals (Q15, F11) → space diagonals in boxes and prisms; more applications (bearings-style navigation, Q13).

Fast-finisher bank — answers

F1. a) $\sqrt{0.81 + 1.44} = \sqrt{2.25} =$ **1.5 m** b) $\sqrt{11.56 - 2.56} = \sqrt{9} =$ **3** c) $\sqrt{5 + 4} = \sqrt{9} =$ **3** d) $\sqrt{10 - 1} =$ **3**

F2. a) $\sqrt{25 \times 2} = 5\sqrt{2}$ b) $\sqrt{36 \times 2} = 6\sqrt{2}$ c) $\sqrt{25 \times 3} = 5\sqrt{3}$ d) $\sqrt{100 \times 2} = 10\sqrt{2}$
e) Q3 d) $\sqrt{75} = 5\sqrt{3}$; Q10 c) $\sqrt{80} = \sqrt{16 \times 5} = 4\sqrt{5}$, $\sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}$.
(Prime factor link: $72 = 2^3 \times 3^2 = (2 \times 3)^2 \times 2$.)

F3. a) $\sqrt{36 + 36} = \sqrt{72} = 6\sqrt{2}$ **cm** (≈ 8.49). b) $s^2 + s^2 = 10^2 \Rightarrow 2s^2 = 100 \Rightarrow$ area $= s^2 =$ **50 cm²**. (In general, area of a square $= \frac{d^2}{2}$ — matches the rhombus/kite formula from Week 9.)

F4. The height bisects the base: $h = \sqrt{10^2 - 5^2} = \sqrt{75} = 5\sqrt{3}$ **cm**. Area $= \frac{1}{2} \times 10 \times 5\sqrt{3} = 25\sqrt{3} \approx$ **43.30 cm²**.

F5. a) Height $= \sqrt{13^2 - 5^2} = 12$; area $= \frac{1}{2} \times 10 \times 12 =$ **60 cm²**.
b) $\frac{1}{2} \times 16 \times h = 120 \Rightarrow h = 15$. Equal sides $= \sqrt{8^2 + 15^2} = 17$. Perimeter $= 17 + 17 + 16 =$ **50 cm**.

F6. Half-diagonals 5 and 12 meet at 90°: side $= 13$ cm. Perimeter **52 cm**; area $= \frac{1}{2} \times 10 \times 24 =$ **120 cm²**.

F7. The perpendicular from the centre bisects the chord. Half-chord $= \sqrt{10^2 - 6^2} = 8$, chord = **16 cm**.

F8. a) $\sqrt{2}, \sqrt{3}, \sqrt{4} = 2, \sqrt{5}$. b) Triangle n has hypotenuse $\sqrt{n + 1}$ (each new triangle adds 1 to the square). $\sqrt{n + 1} = 4 \Rightarrow n + 1 = 16 \Rightarrow$ the **15th** triangle.

F9. $l + w = 17$ and $l^2 + w^2 = 169$. $(l + w)^2 = l^2 + 2lw + w^2 \Rightarrow 289 = 169 + 2lw \Rightarrow lw =$ **60 cm²**. (Check: 5 × 12.)

F10. $PA^2 = x^2 + 4^2$, $PB^2 = (8 - x)^2$. Set equal: $x^2 + 16 = 64 - 16x + x^2 \Rightarrow 16x = 48 \Rightarrow x = 3$. Check: $PA = \sqrt{9 + 16} = 5$, $PB = 8 - 3 = 5$ ✔.

F11. Unfold two adjacent faces into a flat rectangle; the path is its diagonal. Three ways to unfold: $\sqrt{(4 + 3)^2 + 12^2} = \sqrt{193} \approx 13.89$; $\sqrt{(12 + 3)^2 + 4^2} = \sqrt{241} \approx 15.52$; $\sqrt{(12 + 4)^2 + 3^2} = \sqrt{265} \approx 16.28$.
Shortest **≈ 13.89 cm** (cross the 12-cm edge, with the two short dimensions added). It is longer than the 13 cm space diagonal, as it must be — the straight line through the inside is the shortest of all.

F12. a) (3, 4, 5), (6, 8, 10), (5, 12, 13), (9, 12, 15), (8, 15, 17), (12, 16, 20), (7, 24, 25), (15, 20, 25), (10, 24, 26), (20, 21, 29), (18, 24, 30).
b) Products: 60, 480, 780, 1620, 2040, 3840, 4200, 7500, 6240, 12 180, 12 960. Largest common divisor: **60** (in every triple one number is a multiple of 3, one a multiple of 4 and one a multiple of 5).
c) An odd square leaves remainder 1 when divided by 4 ($(2k + 1)^2 = 4k^2 + 4k + 1$); an even square leaves remainder 0 ($(2k)^2 = 4k^2$). If both shorter sides were odd, $a^2 + b^2$ would leave remainder $1 + 1 = 2$ — but no perfect square leaves remainder 2 when divided by 4. So c^2 couldn't be a square. ∎